SentinelRisk AI

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Explainable financial event intelligence and portfolio risk simulation

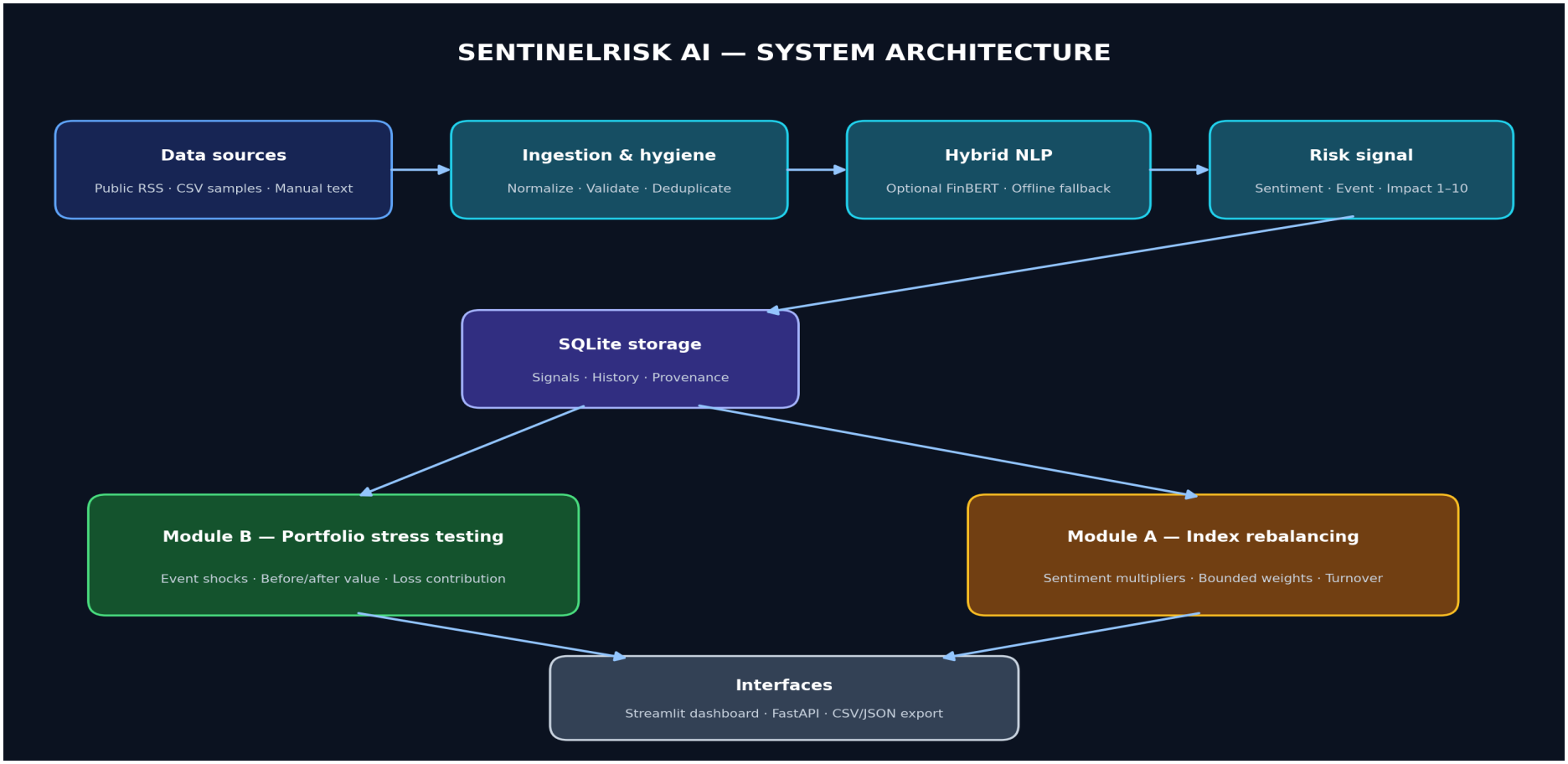
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Prototype · Synthetic data · Explainable scenario modelling

The problem and our approach

- Financial news and social text are unstructured and arrive faster than manual review.
- Risk teams need structured sentiment, event type, severity and affected exposures.
- SentinelRisk normalizes text, scores sentiment, classifies events and emits validated JSON signals.
- One risk engine powers portfolio stress testing and tactical index rebalancing.

Architecture and data flow



Implementation highlights

- Hybrid sentiment: optional FinBERT plus offline lexicon fallback.
- Transparent event taxonomy and weighted impact heuristic with reasons.
- FastAPI endpoints, SQLite persistence, Streamlit dashboard and exports.
- Event-specific synthetic shocks and bounded index weights with validation invariants.

Validation and measurable outputs

- Automated tests cover ingestion, sentiment, events, API and financial invariants.
- Run the demo to generate signals, stress-test results and rebalanced weights.
- Evaluation script uses a small synthetic labelled set.
- Demo metrics are not evidence of real-world performance; independent evaluation is required.

Potential domain impact

- Prioritizes events for analyst review using explainable scores.
- Shows position-level scenario loss contributions.
- Supports what-if analysis with configurable thresholds and scenario assumptions.
- Can later connect to validated exposure data, governance and calibrated models.

Limitations and next steps

- Bundled data and holdings are synthetic; RSS availability varies.
- Keyword classification and lexicon sentiment can miss context.
- Shocks are illustrative, not predictions; instrument valuation is simplified.
- Next: larger labelled evaluation, calibration, entity resolution, backtesting and realistic valuation.